

Real-Time Communication

Task 1. Technical Deep Dive (Otabek, Ogabek)

Explain the architectural differences between **WebRTC**, **WebSockets**, and **SSE (Server-Sent Events)**.

Include: connection model, transport layer usage, latency expectations, and typical use cases.

Task 2. Latency Profiling Experiment (Enkhsod Anujin, Dashdorj Urangoo)

Write a small program (Python, Node.js) that measures end-to-end latency between a client and a local server every 200ms for 10 seconds.

Present results in a short table and interpret the latency distribution.

Task 3. Real-Time System Design (Batsukh Bat Erdene, Nasanjargal Khangaibazar)

Design a real-time system for:

“Live Classroom Streaming + Real-Time Chat + Student Polling”

Your design must include:

- Protocol choice
- Data flow
- Scaling strategy
- Handling high concurrency
- Failure recovery

Task 4. Synchronization Mechanisms (Sayfullo, Jamshid)

Explain how **time synchronization** works in real-time systems.

Discuss: NTP, PTP, clock drift, and why synchronization is critical for multiplayer games or live collaboration.

Task 5. Consistency vs. Real-Time Speed (Batnasan Erdenesambuu, Erdenetsogt Delgerbat)

Write a critical analysis (150–200 words):

“Trade-offs between consistency and real-time performance in distributed systems.”

Task 6. Real-Time Queue Management (Tan Li Xin, Tan Zhi Wei)

Describe how message queues (Kafka, RabbitMQ, Redis Stream) handle real-time workloads.

Compare them based on: throughput, fault tolerance, ordering guarantees, and latency.

Task 7. Handling Unstable Networks (Ozodbek, Jasmina)

Propose **five advanced techniques** to maintain stable real-time communication in environments with fluctuating bandwidth (e.g., mobile networks).

Explain each technique in 3–4 sentences.

Task 8. Packet Loss Simulation (Ganpurev Tengis, Otgonjargal Murun, Enkhsog Anand)

Simulate a scenario where 20% packet loss occurs in a real-time connection.

Write a short explanation of how:

- TCP
- UDP
- WebRTC

react to packet loss and which is best for real-time video/audio.

Task 9. Protocol Extension Task (Lui Yanzhao)

Choose one protocol (RTP, SCTP, or QUIC).

Explain its architecture deeply and propose **one improvement** that could reduce latency or packet loss in real-time scenarios.